
TN0: CANONICAL NOTATION AND CONVENTIONS FOR CONSTRAINT-SURFACE DYNAMICS

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ABSTRACT

This technical note establishes canonical notation and conventions for the Constraint-Surface Dynamics (CSD) programme. All CSD papers after Paper D adopt these conventions. Where earlier papers (A through D) use different notation, the differences are stated explicitly. Future consolidated editions of those papers will align to the present standard.

Contents

1	Differential Geometry Preliminaries	3
1.1	Smooth manifolds and maps	3
1.2	Differential forms and operations	3
1.3	Integration and measures	3
1.4	Inner product conventions	3
1.5	Index conventions	3
2	The Ontic Layer	4
2.1	The ontic manifold	4
2.2	Ontic microstate and trajectory	4
2.3	Symplectic and Kähler structure	4
2.4	Liouville measure on Σ	4
2.5	Ontic Hamiltonian and flow	4
2.6	Fibres	4
2.7	Outcome regions and context dependence	5
3	The Epistemic Layer	5
3.1	Epistemic state space and dimension convention	5
3.2	Epistemic projection	5
3.3	Fubini–Study geometry and measure	5
3.4	Epistemic density as a Radon–Nikodym derivative	5
3.5	Voronoi partitions on $\mathbb{CP}^{N-1}$	6

3.6	Epistemic Hamiltonian flow	6
3.7	Operators and hat convention	6
3.8	Density operators	6
3.9	Projectors, POVMs, and Probabilities	6
3.9.1	Projectors and POVMs	6
3.9.2	Epistemic outcome regions	7
3.9.3	Probability from an epistemic state	7
3.9.4	Sharp preparation idealisation	7
3.9.5	Operator-form probabilities	8
3.9.6	Correction note: state-independent ratios are not general	8
4	Truncated Systems	8
4.1	Truncation conventions	8
4.2	Energy basis and populations	8
4.3	Phase conventions	9
4.4	Invariant sets	9
4.5	Moment map (corrected symbol)	9
5	Composite Systems	9
5.1	Bipartite structure	9
5.2	System–apparatus and environment	10
5.3	Entanglement and marginalisation	10
6	Physical Systems and Constants	10
6.1	Fundamental constants	10
6.2	Harmonic oscillator conventions	10
6.3	Pointer and apparatus variables (TN3 onward)	10
7	Preparations	10
7.1	Ontic preparations	10
7.2	Epistemic preparations	11
7.3	Associated density operator	11
8	Bell Scenarios and Correlations	11
8.1	Detector settings and outcomes	11
8.2	Correlations	11
8.3	Context-indexed outcome maps	11
9	Visibility and Isolation	11
9.1	Visibility	11
9.2	Isolation (dynamical separation)	12
9.3	Status of the visibility-isolation relation	12

10 Programme-Level Conventions	12
10.1 Paper labelling	12
10.2 Cross-reference format	12
10.3 Axiom numbering	12

Introduction

CSD uses a multi-layer description: an ontic manifold Σ supporting deterministic Hamiltonian flow and a Liouville measure, and an epistemic projective state space $\mathbb{CP}^{N-1}$ obtained via a smooth many-to-one projection $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$. Because CSD spans symplectic geometry, projective geometry, and operational quantum notation, symbol collisions are otherwise unavoidable. TN0 fixes: (i) canonical symbols and index conventions, (ii) measure and pushforward notation, (iii) separation of ontic, epistemic, and operator layers, and (iv) standardised composite-system subscripts. Minor mathematical hygiene points are enforced explicitly, including completeness of smooth vector fields on compact manifolds and Radon–Nikodym notation for densities.

1 Differential Geometry Preliminaries

1.1 Smooth manifolds and maps

All manifolds in this programme are smooth, finite-dimensional, and connected unless otherwise stated. For a smooth manifold M , $C^\infty(M)$ denotes the ring of smooth real-valued functions. If $f : M \rightarrow N$ is a smooth map, then f_* denotes the pushforward on tangent vectors and on measures where appropriate, and f^* denotes the pullback on differential forms.

1.2 Differential forms and operations

d denotes the exterior derivative. For a vector field X on M : ι_X denotes contraction (interior product) and $\mathcal{L}_X$ denotes the Lie derivative. The Cartan identity $\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$ is used without further comment. $\Omega^p(M)$ is the space of smooth p -forms and $\wedge$ denotes the wedge product.

1.3 Integration and measures

If M is orientable of dimension m , a volume form is a nowhere-vanishing m -form. The Borel measure associated with a volume form η is written μ_η or given a specific name (μ_L , μ_{FS}). Compactness implies finiteness of total measure; normalisation conventions are stated where used.

1.4 Inner product conventions

Dirac notation $\langle \psi | \phi \rangle$ is used for the Hermitian inner product on $\mathbb{C}^N$ in the main text. The mathematician notation $\langle \psi, \phi \rangle$ is permitted in appendices. They are interchangeable:

$$\langle \psi | \phi \rangle \equiv \langle \psi, \phi \rangle,$$

with physics convention: conjugate-linear in the first argument and linear in the second.

1.5 Index conventions

Lower Latin indices $a, b, c, \dots$ denote real coordinates on smooth manifolds. Complex holomorphic indices $k, \ell = 1, \dots, N-1$ are used in affine charts on $\mathbb{CP}^{N-1}$; anti-holomorphic indices are $\bar{k}, \bar{\ell}$. Lower Greek indices $\mu, \nu, \dots$ are reserved for emergent spacetime coordinates in future extensions (Paper Σ onward) and do not appear in finite-dimensional papers. Einstein summation is used only when stated; otherwise sums are written explicitly.

2 The Ontic Layer

2.1 The ontic manifold

Σ denotes the ontic manifold: a compact, connected, finite-dimensional Kähler manifold

$$(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma), \quad g_\Sigma(u, v) = \omega_\Sigma(u, J_\Sigma v).$$

Subsystem labels: Σ_A, Σ_B denote individual subsystems; Σ_{AB} denotes a joint ontic manifold. In general $\Sigma_{AB} \neq \Sigma_A \times \Sigma_B$.

Truncations: Σ_N denotes the ontic manifold associated with an N -dimensional truncation. In the standard Hopf construction, $\Sigma_N = S^{2N-1}$.

2.2 Ontic microstate and trajectory

The ontic microstate is a point $x \in \Sigma$. Its ontic trajectory is $x(t) \in \Sigma$. The symbol ω is reserved for symplectic forms (always subscripted) and for angular frequency (unsubscripted).

2.3 Symplectic and Kähler structure

ω_Σ is the symplectic form on Σ ; ω_{FS} is the Fubini–Study symplectic form on $\mathbb{CP}^{N-1}$. The bare symbol ω never denotes a symplectic form. Compatibility is recorded in the standard Kähler sense:

$$g_\Sigma(u, v) = \omega_\Sigma(u, J_\Sigma v), \quad \nabla^{LC} J_\Sigma = 0,$$

where ∇^{LC} is the Levi–Civita connection of g_Σ .

The full Kähler structure of Σ , including the explicit compatibility relations among ω_Σ , J_Σ , and g_Σ , is specified in the paper where Σ is constructed in detail. TN0 records only the canonical notation and the load-bearing convention that these structures are taken to be mutually compatible.

2.4 Liouville measure on Σ

Let $\dim_{\mathbb{R}} \Sigma = 2d$. The Liouville volume form is

$$\Omega_\Sigma = \frac{\omega_\Sigma^d}{d!},$$

and the associated Liouville measure is

$$\mu_L(U) = \int_U \Omega_\Sigma, \quad U \subset \Sigma \text{ measurable.}$$

By compactness, $\mu_L(\Sigma) < \infty$. The invariant measure is always written μ_L , never bare μ .

2.5 Ontic Hamiltonian and flow

$H : \Sigma \rightarrow \mathbb{R}$ is a smooth Hamiltonian. The Hamiltonian vector field X_H is defined by

$$\iota_{X_H} \omega_\Sigma = dH.$$

The ontic Hamiltonian flow is $\Phi_t : \Sigma \rightarrow \Sigma$ satisfying

$$\frac{d}{dt} \Phi_t(x) = X_H(\Phi_t(x)), \quad \Phi_0 = \text{id}.$$

Since Σ is compact and X_H is smooth, X_H is complete, hence Φ_t exists for all $t \in \mathbb{R}$. The flow preserves Liouville measure:

$$\mu_L(\Phi_t(U)) = \mu_L(U)$$

for all measurable $U \subset \Sigma$.

2.6 Fibres

For $[\psi] \in \mathbb{CP}^{N-1}$, the fibre is

$$F_{[\psi]} = \pi^{-1}([\psi]) \subset \Sigma.$$

In the Hopf example $\Sigma_N = S^{2N-1}$, fibres are circles S^1 .

2.7 Outcome regions and context dependence

$\Omega_i^\Sigma \subset \Sigma$ denotes the ontic outcome region for outcome i in a given measurement context. The regions $\{\Omega_i^\Sigma\}$ form a measurable partition of the relevant preparation region up to μ_L -null sets, and may depend on the measurement context.

Epistemic outcome regions are subsets $\Omega_i^{ep} \subset \mathbb{CP}^{N-1}$. When needed, write $\Omega_i^{ep}(M)$ to emphasise dependence on the measurement context M .

3 The Epistemic Layer

3.1 Epistemic state space and dimension convention

$\mathbb{CP}^{N-1}$ is complex projective space of rays in $\mathbb{C}^N$. Uppercase N always denotes Hilbert-space dimension or truncation cutoff; lowercase n is reserved for quantum numbers and summation indices.

An epistemic state is $[\psi] \in \mathbb{CP}^{N-1}$. A representative vector is $|\psi\rangle \in \mathbb{C}^N$. The unit sphere is $S^{2N-1} = \{|\psi\rangle \in \mathbb{C}^N : \langle\psi|\psi\rangle = 1\}$.

3.2 Epistemic projection

$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$ is the epistemic projection, assumed smooth, surjective, and many-to-one. In the truncated Hopf-model presentation, one writes

$$\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1}, \quad \pi_N(|\psi\rangle) = [\psi].$$

The *quantum-effective sector* is the regime identified in Paper C, Axiom A5 and Appendix I: a region $S \subset \Sigma$ together with an operational timescale $[0, T]$ for which the ontic Hamiltonian admits a decomposition

$$H = h \circ \pi + \delta H,$$

with fibre variation uniformly bounded by a small parameter ϵ , the ontic trajectory remains in S on $[0, T]$, the accumulated projection error satisfies $K_T \epsilon \ll 1$, and the experimentally relevant observables depend only on the projected epistemic state to the relevant operational resolution.

In the symmetry-compatible quantum-effective sector, and under the $SU(N)$ -equivariance assumptions of TN1, the pushforward measure satisfies

$$\pi_* \mu_L = c \mu_{FS}, \quad c > 0,$$

where c is a *global constant* on $\mathbb{CP}^{N-1}$, not a state-dependent density. Its constancy is established in TN1 from $SU(N)$ -equivariance and transitivity.

3.3 Fubini–Study geometry and measure

g_{FS} is the Fubini–Study metric. The geodesic distance is

$$d_{FS}([\psi], [\phi]) = \arccos \left(\frac{|\langle\psi|\phi\rangle|}{\|\psi\| \|\phi\|} \right).$$

ω_{FS} is the associated symplectic form. The Fubini–Study volume form is

$$\Omega_{FS} = \frac{\omega_{FS}^{N-1}}{(N-1)!},$$

and the associated measure is μ_{FS} . When a probability measure is required, use $\mu_{FS}/\mu_{FS}(\mathbb{CP}^{N-1})$.

3.4 Epistemic density as a Radon–Nikodym derivative

Given a preparation measure μ_{prep} on Σ , define the pushforward $\pi_* \mu_{prep}$ on $\mathbb{CP}^{N-1}$. If $\pi_* \mu_{prep} \ll \mu_{FS}$, define the epistemic density by the Radon–Nikodym derivative

$$\rho_{ep} := \frac{d(\pi_* \mu_{prep})}{d\mu_{FS}}, \quad \text{so that} \quad \pi_* \mu_{prep} = \rho_{ep} \mu_{FS}.$$

The subscript “ep” distinguishes ρ_{ep} from the density operator $\hat{\rho}$.

3.5 Voronoi partitions on $\mathbb{CP}^{N-1}$

A projective measurement context M with outcomes indexed by i selects distinct reference rays $\{[\phi_i]\} \subset \mathbb{CP}^{N-1}$. The Fubini–Study Voronoi cells are

$$\Omega_i^{ep}(M) = \left\{ [\psi] \in \mathbb{CP}^{N-1} : d_{FS}([\psi], [\phi_i]) < d_{FS}([\psi], [\phi_j]) \forall j \neq i \right\}.$$

For generic contexts, boundaries are codimension-1 smooth hypersurfaces away from a lower-dimensional singular locus, and are μ_{FS} -null.

3.6 Epistemic Hamiltonian flow

$h : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}$ is the epistemic Hamiltonian. The vector field X_h satisfies $\iota_{X_h} \omega_{FS} = dh$, and the epistemic flow is $\Psi_t : \mathbb{CP}^{N-1} \rightarrow \mathbb{CP}^{N-1}$. In the exact quantum-effective limit, the ontic Hamiltonian satisfies

$$H = h \circ \pi,$$

for some smooth function $h : CP^{N-1} \rightarrow \mathbb{R}$. In that ideal case the induced epistemic flow is denoted

$$\Psi_t : CP^{N-1} \rightarrow CP^{N-1},$$

with

$$\pi \circ \Phi_t = \Psi_t \circ \pi.$$

More generally, the current CSD programme allows an approximate-projectability regime, in which projected closure is controlled rather than exact.

3.7 Operators and hat convention

All operators on $\mathbb{C}^N$ carry a hat: $\hat{H}, \hat{A}, \hat{x}, \hat{p}, \hat{n}, \hat{a}, \hat{a}^\dagger$. The Ashtekar–Schilling correspondence is

$$h([\psi]) = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}.$$

Schrödinger sign convention:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle.$$

Some mathematical treatments use the opposite sign convention after moving the factor of i to the other side; no substantive difference is intended.

3.8 Density operators

$\hat{\rho}$ is a positive semidefinite, trace-one operator: $\hat{\rho} \geq 0$, $\text{Tr}[\hat{\rho}] = 1$. In CSD, $\hat{\rho}$ is derived (TN4), not primitive.

3.9 Projectors, POVMs, and Probabilities

This section fixes the canonical probability conventions used throughout CSD. It replaces earlier state-independent formulas and separates three levels of description: (i) epistemic measures on $\mathbb{CP}^{N-1}$, (ii) sharp preparation as an idealisation, and (iii) the operator (trace) form used in standard quantum mechanics. Notation follows TN0 throughout.

3.9.1 Projectors and POVMs

Let $\mathcal{H}_N \cong \mathbb{C}^N$ and let $\hat{\mathbb{I}}$ denote the identity operator on $\mathcal{H}_N$.

Rank-one projectors: Given an orthonormal basis $\{|e_i\rangle\}_{i=1}^N$, define

$$\Pi_i := |e_i\rangle \langle e_i|, \quad \sum_{i=1}^N \Pi_i = \hat{\mathbb{I}}.$$

POVM elements: A POVM is a collection of operators $\{F_k\}_{k \in K}$ on $\mathcal{H}_N$ satisfying

$$F_k \succeq 0, \quad \sum_{k \in K} F_k = \hat{\mathbb{I}}.$$

3.9.2 Epistemic outcome regions

Let M denote a fixed measurement context on the truncated epistemic space

$$\mathbb{CP}^{N-1}.$$

An *epistemic outcome partition* for M is a family of Borel sets

$$\{\Omega_i^{\text{ep}}(M)\}_{i \in I} \subset \mathcal{B}(\mathbb{CP}^{N-1})$$

such that

$$\mu_{FS}\left(\mathbb{CP}^{N-1} \setminus \bigcup_{i \in I} \Omega_i^{\text{ep}}(M)\right) = 0,$$

and

$$\mu_{FS}(\Omega_i^{\text{ep}}(M) \cap \Omega_j^{\text{ep}}(M)) = 0 \quad (i \neq j).$$

For each i , the boundary $\partial\Omega_i^{\text{ep}}(M)$ is required to satisfy

$$\mu_{FS}(\partial\Omega_i^{\text{ep}}(M)) = 0.$$

Accordingly, the family $\{\Omega_i^{\text{ep}}(M)\}_{i \in I}$ defines a partition of $\mathbb{CP}^{N-1}$ up to μ_{FS} -null sets.

In the projective case, the index set I coincides with the projector labels. In the POVM case, one may write

$$\{\Omega_k^{\text{ep}}(M)\}_{k \in K}$$

to match the POVM labels.

Axiom A7 requires only the existence of such context-dependent measurable outcome regions with μ_{FS} -null boundaries. Fubini–Study Voronoi regions provide a canonical finite-dimensional projective exemplar, but are not assumed to be universal and are not part of the abstract statement of A7.

3.9.3 Probability from an epistemic state

An epistemic state is represented by a Borel probability measure ν_{ep} on $\mathbb{CP}^{N-1}$. If ν_{ep} is absolutely continuous with respect to μ_{FS} , write

$$\nu_{ep} \ll \mu_{FS}, \quad \rho_{ep} := \frac{d\nu_{ep}}{d\mu_{FS}}.$$

Then the probability of outcome i in context M is

$$P(i \mid \nu_{ep}, M) := \nu_{ep}(\Omega_i^{\text{ep}}(M)) = \int_{\Omega_i^{\text{ep}}(M)} \rho_{ep} d\mu_{FS}. \quad (1)$$

If $\nu^{\text{ep}} \ll \mu_{FS}$, then modifications of $\Omega_i^{\text{ep}}(M)$ on μ_{FS} -null sets do not change the probability $P(i \mid \nu^{\text{ep}}, M)$.

3.9.4 Sharp preparation idealisation

In many arguments, a sharply prepared pure state at ray $[\psi] \in \mathbb{CP}^{N-1}$ is treated as an idealisation. In CSD this is handled in one of two equivalent ways, depending on context.

(A) Operator form (preferred canonical idealisation): A sharply prepared pure state corresponds to the rank-one density operator

$$\hat{\rho}_{[\psi]} := |\psi\rangle\langle\psi|.$$

Probabilities are then computed using the operator rules in Section 3.9.5 below.

(B) Localisation limit (measure-theoretic idealisation): Alternatively, represent sharp preparation by a family of absolutely continuous measures $\nu_{ep}^{(r)}$ concentrated near $[\psi]$, for example by restricting and renormalising μ_{FS} (or $\rho_{ep}\mu_{FS}$) to Fubini–Study balls $B_r([\psi])$ and letting $r \rightarrow 0$.

When limits are used, they are understood in the standard differentiation sense: for measurable regions whose boundaries are μ_{FS} -null, ratios of the form

$$\frac{\mu_{FS}(\Omega_i^{ep}(M) \cap B_r([\psi]))}{\mu_{FS}(B_r([\psi]))}$$

converge for μ_{FS} -almost every $[\psi]$. The load-bearing regularity condition here is the μ_{FS} -null boundary assumption, which is the condition needed for the differentiation theorem to apply in the present metric-measure setting. In the main text, the operator idealisation (A) is the default.

See, for example, standard differentiation theorems for finite Borel measures on metric spaces.

3.9.5 Operator-form probabilities

Given a density operator $\hat{\rho}$ on $\mathcal{H}_N$ and a measurement context M :

Projective measurement: If M is associated to projectors $\{\Pi_i\}_{i \in I}$, then

$$P(i \mid \hat{\rho}, M) = \text{Tr}(\hat{\rho} \Pi_i). \quad (2)$$

POVM measurement: If M is associated to a POVM $\{F_k\}_{k \in K}$, then

$$P(k \mid \hat{\rho}, M) = \text{Tr}(\hat{\rho} F_k). \quad (3)$$

3.9.6 Correction note: state-independent ratios are not general

Earlier drafts sometimes used a state-independent ratio of the form

$$\frac{\mu_{FS}(\Omega_i^{ep}(M))}{\mu_{FS}(\mathbb{CP}^{N-1})}.$$

This expression corresponds only to the *uniform* epistemic state (constant density with respect to μ_{FS}). In general, probabilities depend on the epistemic state via (1) (or equivalently the operator rule (2)–(3) when the epistemic state is represented by $\hat{\rho}$). Accordingly, state-independent ratios must not be used as the general probability rule.

4 Truncated Systems

4.1 Truncation conventions

When a continuous-variable system is modelled in a finite-dimensional subspace, subscript N denotes truncated objects:

$$\begin{aligned} \mathcal{H}_N &= \text{span}\{|0\rangle, \dots, |N-1\rangle\}, \\ \mathbb{CP}^{N-1} &\text{ is the truncated epistemic state space,} \\ \Sigma_N &= S^{2N-1} \text{ in the standard Hopf construction,} \\ \pi_N : \Sigma_N &\rightarrow \mathbb{CP}^{N-1} \text{ the Hopf fibration, } \pi_N(|\psi\rangle) = [\psi], \\ \hat{H}_N &= \hat{H}|_{\mathcal{H}_N}, H_N([\psi]) = \frac{\langle \psi | \hat{H}_N | \psi \rangle}{\langle \psi | \psi \rangle}, \\ \tilde{H}_N &:= H_N \circ \pi_N \text{ (lift to } \Sigma_N; \text{ tilde always denotes such a lift).} \end{aligned} \quad (4)$$

If both finite- N and limiting objects appear in one discussion, the subscript is mandatory.

4.2 Energy basis and populations

$|n\rangle$ denotes an energy eigenstate of $\hat{H}$ (or $\hat{H}_N$) with eigenvalue E_n , for $n = 0, 1, \dots, N-1$:

$$\hat{H} |n\rangle = E_n |n\rangle.$$

Amplitudes and populations in the energy basis:

$$c_n := \langle n | \psi \rangle, \quad p_n := |c_n|^2, \quad \sum_{n=0}^{N-1} p_n = 1.$$

The population vector $p = (p_0, \dots, p_{N-1})$ lies in the simplex Δ_{N-1} .

4.3 Phase conventions

On S^{2N-1} write $c_n = r_n e^{i\theta_n}$ with $r_n \geq 0$. Relative phases are

$$\varphi_k := \theta_k - \theta_0, \quad k = 1, \dots, N-1.$$

For diagonal Hamiltonians in the energy basis, populations are constant and

$$\dot{\varphi}_k = -\frac{E_k - E_0}{\hbar}.$$

4.4 Invariant sets

For a population vector p , T_p denotes the invariant torus in $\mathbb{CP}^{N-1}$ associated with p . Generically (all $p_n > 0$), T_p is a real torus of dimension $N-1$.

For the eigenstate ray $[|n\rangle]$, the fibre circle in Σ_N is

$$T_n^{(1)} = \{e^{i\alpha} |n\rangle : \alpha \in [0, 2\pi)\} \subset \Sigma_N.$$

4.5 Moment map (corrected symbol)

The moment map for the maximal torus action (relative to the standard diagonal $U(1)^{N-1}$ action) is

$$m : \mathbb{CP}^{N-1} \rightarrow \Delta_{N-1}, \quad m([\psi]) = (p_0, \dots, p_{N-1}).$$

The letter m is used to avoid collision with the measure symbols μ_L and μ_{FS} .

5 Composite Systems

5.1 Bipartite structure

Let Σ_{AB} denote the joint ontic manifold for a bipartite system AB . In general,

$$\Sigma_{AB} \neq \Sigma_A \times \Sigma_B. \quad (5)$$

This non-factorisation is a structural statement about the ontic geometry in the present CSD architecture, where the joint ontic manifold is not taken to be a Cartesian product of subsystem ontic manifolds.

It is not a claim of operational signalling. Operational no-signalling is enforced at the epistemic level by the existence of well-defined marginals (See TN4).

Dimensions and epistemic space: Let $\dim \mathcal{H}_A = N_A$ and $\dim \mathcal{H}_B = N_B$. The joint epistemic space is

$$\mathbb{CP}^{N_A N_B - 1},$$

with projection

$$\pi_{AB} : \Sigma_{AB} \rightarrow \mathbb{CP}^{N_A N_B - 1}.$$

Product epistemic description (when applicable): When an *approximate product epistemic description* is appropriate (for example, for independently prepared subsystems or in regimes where correlations are negligible), we use the product ray space

$$\mathbb{CP}^{N_A - 1} \times \mathbb{CP}^{N_B - 1}.$$

In that case, we represent the epistemic description by a map

$$\pi_{AB}^\times : \Sigma_{AB} \rightarrow \mathbb{CP}^{N_A - 1} \times \mathbb{CP}^{N_B - 1},$$

and define marginal projections by composition with the canonical product projections

$$\pi_A := \text{pr}_A \circ \pi_{AB}^\times, \quad \pi_B := \text{pr}_B \circ \pi_{AB}^\times, \quad (6)$$

where pr_A, pr_B are the projections from $\mathbb{CP}^{N_A - 1} \times \mathbb{CP}^{N_B - 1}$ onto its factors.

Remark (avoidance of an incorrect factorisation claim): Even when a product epistemic description is used, (5) remains in force at the ontic level. The product construction concerns the *epistemic representation* and the availability of marginals, not an ontic Cartesian product structure.

5.2 System–apparatus and environment

Σ_{SA} denotes system–apparatus ontology, Σ_{SAE} denotes system–apparatus–environment ontology: Subscript ordering is always system first, then apparatus, then environment.

5.3 Entanglement and marginalisation

$\hat{\rho}_{AB}$ denotes a joint density operator on $\mathbb{C}^{N_A} \otimes \mathbb{C}^{N_B}$. The reduced density operator is

$$\hat{\rho}_A := \text{Tr}_B(\hat{\rho}_{AB}).$$

When a product epistemic description, or other admissible marginal map, has been specified, marginals are defined by pushforward of measures. If $\rho_{ep,AB}$ is a density on the joint epistemic space with respect to the appropriate Fubini–Study measure, define the marginal density $\rho_{ep,A}$ by

$$(\pi_A)_* \mu_{prep,AB} = \rho_{ep,A} \mu_{FS,A},$$

where $\mu_{FS,A}$ is the Fubini–Study measure on $\mathbb{CP}^{N_A-1}$ and $\mu_{prep,AB}$ is the composite preparation measure on Σ_{AB} .

6 Physical Systems and Constants

6.1 Fundamental constants

$\hbar$ denotes the reduced Planck constant and is written explicitly unless otherwise stated. The convention $\hbar = 1$ is used only when explicitly declared at the opening of a calculation.

ω (without subscript) denotes angular frequency in physical expressions. Symplectic forms are always subscripted: $\omega_\Sigma, \omega_{FS}$.

6.2 Harmonic oscillator conventions

For a harmonic oscillator with mass m and angular frequency ω :

$$[\hat{a}, \hat{a}^\dagger] = 1, \quad \hat{n} = \hat{a}^\dagger \hat{a}, \quad \hat{H} = \hbar\omega \left(\hat{n} + \frac{1}{2} \right).$$

Define the characteristic length $\ell_0 = \sqrt{\hbar/(m\omega)}$ and

$$\hat{x} = \frac{\ell_0}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = \frac{i\hbar}{\sqrt{2}\ell_0}(\hat{a}^\dagger - \hat{a}).$$

6.3 Pointer and apparatus variables (TN3 onward)

q_P and p_P denote pointer position and momentum. Subscript P always denotes pointer degrees of freedom. κ denotes the pointer coupling constant in readout Hamiltonians.

7 Preparations

7.1 Ontic preparations

A preparation defines a probability measure on Σ absolutely continuous with respect to μ_L :

$$d\mu_{prep}(x) = f(x) d\mu_L(x),$$

where $f \in L^1(\Sigma, \mu_L)$, $f \geq 0$, and $\int_\Sigma f d\mu_L = 1$.

Pure preparations may be represented, at the ontic level, by sharply localised densities supported in neighbourhoods of fibres $F_{[\psi]}$.

7.2 Epistemic preparations

The epistemic preparation measure is the pushforward $\pi_*\mu_{prep}$ on $\mathbb{CP}^{N-1}$. If $\pi_*\mu_{prep} \ll \mu_{FS}$, define $\rho_{ep,prep} = d(\pi_*\mu_{prep})/d\mu_{FS}$ so that

$$\pi_*\mu_{prep} = \rho_{ep,prep} \mu_{FS}.$$

When the preparation is clear from context, the subscript “prep” may be dropped.

7.3 Associated density operator

Given $\rho_{ep,prep}$, the associated density operator $\hat{\rho}_{prep}$ is defined by the representation (TN4) such that for all projectors Π corresponding to operational outcomes,

$$\text{Tr}(\hat{\rho}_{prep}\Pi) = \int_{\Omega^{ep}(\Pi)} \rho_{ep,prep} d\mu_{FS},$$

where $\Omega^{ep}(\Pi)$ is the epistemic outcome region associated with Π in the relevant context. This representation is derived, not postulated.

8 Bell Scenarios and Correlations

8.1 Detector settings and outcomes

In bipartite Bell experiments:

$$a, b \in S^2 \text{ are detector-setting vectors, } s, t \in \{+1, -1\} \text{ are outcomes.}$$

Let $\theta = \arccos(a \cdot b)$ denote the relative angle.

8.2 Correlations

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P(s, t | a, b).$$

The CHSH quantity is

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b'),$$

with Bell bound $|S| \leq 2$ and quantum maximum $|S| = 2\sqrt{2}$.

8.3 Context-indexed outcome maps

For a given context (a, b) , the deterministic outcome map is

$$F^{(a,b)} : \Sigma_{AB} \rightarrow \{+1, -1\} \times \{+1, -1\}.$$

If Bell inequalities are violated, there is no single measurable global map assigning outcomes for all incompatible settings simultaneously (c1, Theorem 3.1).

9 Visibility and Isolation

9.1 Visibility

V denotes interference visibility, defined operationally by

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}},$$

where $I_{\max}$ and $I_{\min}$ are measured intensities. Here I denotes intensity, not isolation.

9.2 Isolation (dynamical separation)

I denotes the dynamical isolation parameter, defined to be dimensionless and constrained to

$$I \in [0, 1].$$

$I = 0$ corresponds to full coherence (maximal overlap of relevant outcome images) and $I \rightarrow 1$ corresponds to macroscopic separation.

Operational forms used in applications include:

$$I(\tau) \approx 1 - e^{-\Gamma\tau} \text{ (decoherence-dominated),} \quad I(g) \approx g_{\text{eff}}/g_{\text{max}} \text{ (tunable-interaction regimes).}$$

9.3 Status of the visibility-isolation relation

A central proposed empirical relation in CSD is

$$V \approx 1 - I.$$

At present this relation has not been derived from ontic dynamics in full generality. A derivation with explicit error bounds, together with clearly stated restrictions on the Hamiltonian class and regime of validity, remains an open problem in the programme. Accordingly, TN0 records V and I as canonical definitions, but does not treat the relation $V \approx 1 - I$ as an exact theorem.

10 Programme-Level Conventions

10.1 Paper labelling

Papers A–D denote the core foundation papers. TN0–TN7 denote technical notes. c1–c3 denote companion or bridge papers. QI papers denote the quantum-information branch. Application papers are treated as a separate class.

10.2 Cross-reference format

Internal citations use the format: (Paper C, §3.2), (TN1, Theorem 3.1), (TN3, Appendix C). External references use standard bibliographic conventions.

10.3 Axiom numbering

The axiom system for Σ is defined in Paper C, Appendix A as Axioms A1 through A7. For programme-level auditing, A4 carries an internal distinction between:

- A4a: the minimal measurability and finite-pushforward assumptions needed to state epistemic probabilities and densities;
- A4b: the sector-conditional measure bridge

$$\pi_*\mu_L = c\mu_{FS},$$

established in the symmetry-compatible quantum-effective sector via the $SU(N)$ -equivariance route of TN1.

This internal split is clarificatory and does not renumber the canonical Paper C axiom list. If further axioms are introduced in future work, they are appended as A8, A9, ... rather than renumbered.

Appendix: Consolidated Symbol Table

Symbol	Meaning	First defined	Supersedes
Σ	Ontic manifold	Paper A	—
Σ_N	Truncated ontic manifold ($= S^{2N-1}$ in Hopf case)	TN5	—
Σ_{AB}	Joint ontic manifold	Paper C	—
Σ_{SA}	System–apparatus ontic manifold	TN3	—
Σ_{SAE}	System–apparatus–environment ontic manifold	Paper D	—
x	Ontic microstate (point on Σ)	Paper C	ω in Papers A, D
$x(t)$	Ontic trajectory	Paper C	$\omega(t)$ in Papers A, D
ω_Σ	Symplectic form on Σ	Paper C	—
ω_{FS}	Fubini–Study symplectic form on $\mathbb{CP}^{N-1}$	Paper B	—
ω	Angular frequency (unsubscripted only)	TN5	—
J_Σ	Complex structure on Σ	Paper C	—
g_Σ	Riemannian metric on Σ	Paper C	—
g_{FS}	Fubini–Study metric on $\mathbb{CP}^{N-1}$	Paper B	—
d_{FS}	Fubini–Study geodesic distance	Paper B	—
μ_L	Liouville measure on Σ	TN0	bare μ in Paper A; μ_Σ in Paper D
μ_{FS}	Fubini–Study measure on $\mathbb{CP}^{N-1}$	Paper B	—
Ω_Σ	Liouville volume form on Σ	TN0	—
Ω_{FS}	Fubini–Study volume form on $\mathbb{CP}^{N-1}$	TN0	—
Ω_i^Σ	Ontic outcome region on Σ (context-dependent)	TN0	—
Ω_i^{ep}	Epistemic outcome region on $\mathbb{CP}^{N-1}$	Paper C	—
Ω_0	Pre-measurement preparation region	Paper D	—
H	Ontic Hamiltonian $H : \Sigma \rightarrow \mathbb{R}$	Paper C	—
h	Epistemic Hamiltonian $h : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}$	Paper C	—
$\tilde{H}_N$	Lifted Hamiltonian on $\Sigma_N (= H_N \circ \pi_N)$	TN5	—
$\hat{H}$	Hamiltonian operator on $\mathbb{C}^N$	Paper B	—
$\hat{H}_N$	Truncated Hamiltonian operator	TN5	—
X_H	Hamiltonian vector field on Σ	TN0	—
X_h	Hamiltonian vector field on $\mathbb{CP}^{N-1}$	TN0	—
Φ_t	Ontic Hamiltonian flow on Σ	TN0	$\varphi(t)$ in Paper A
Ψ_t	Epistemic Hamiltonian flow on $\mathbb{CP}^{N-1}$	TN0	Φ_t on $\mathbb{CP}^{N-1}$ in TN5
π	Epistemic projection $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$	Paper C	—
π_N	Hopf fibration $\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1}$	TN5	—
π_{AB}	Joint projection for composite system	Paper C	—
π_A, π_B	Marginal projections	Paper C	—
$F^{[\psi]}$	Fibre over $[\psi]$	Paper C	—
$\mathbb{CP}^{N-1}$	Epistemic state space (projective space)	TN0	$\mathbb{CP}^{n-1}$ in Papers B, C
$[\psi]$	Epistemic state (ray)	Paper B	—
$ \psi\rangle$	Representative vector of ray $[\psi]$	Paper B	—
ρ_{ep}	Epistemic density on $\mathbb{CP}^{N-1}$	TN0	ρ in Paper C
$\hat{\rho}$	Density operator on $\mathbb{C}^N$	TN4	—
$\hat{\rho}_{prep}$	Density operator for a preparation	TN4	—
$\hat{\rho}_A$	Reduced density operator for subsystem A	TN4	—
Π_i	Rank-one projector $\Pi_i = e_i\rangle\langle e_i $	Paper B	—
F_k	POVM element	TN0	E_k in TN4
$ n\rangle$	Energy eigenstate	Paper B	—
E_n	Energy eigenvalue	Paper B	—
c_n	$c_n = \langle n \psi\rangle$	TN5	—
p_n	Population $p_n = c_n ^2$	TN5	—
T_p	Invariant torus for population vector p	TN5	—
$T_n^{(1)}$	$U(1)$ fibre circle above $[n\rangle]$ in Σ_N	TN0	—
m	Moment map $m : \mathbb{CP}^{N-1} \rightarrow \Delta_{N-1}$	TN0	μ in TN5
μ_{prep}	Preparation measure on Σ	TN0	—
$\hat{a}, \hat{a}^\dagger$	Annihilation and creation operators	TN5	—
$\hat{n}$	Number operator $\hat{a}^\dagger \hat{a}$	TN5	—
$\hat{x}$	Position operator	TN5	—
$\hat{x}_N$	Truncated position operator	TN5	—
q_P, p_P	Pointer position and momentum	TN3	—
κ	Pointer coupling constant	TN3	—
a, b	Detector setting vectors (Bell)	TN2	—
s, t	Binary outcomes $\in \{\pm 1\}$	TN2	—
$E(a, b)$	Two-party correlation function	TN2	—
S	CHSH quantity	TN2	—
$F^{(a,b)}$	Context-indexed outcome map	c1	—
V	Interference visibility	Paper D	—

Quick-Reference Card

Symbol	One-line meaning	§Ref
$x \in \Sigma$	Ontic microstate on the constraint surface	2.2
$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$	Epistemic projection (many-to-one)	3.2
$\Phi_t : \Sigma \rightarrow \Sigma$	Ontic Hamiltonian flow (μ_L -preserving)	2.5
$\Psi_t : \mathbb{CP}^{N-1} \rightarrow \mathbb{CP}^{N-1}$	Epistemic Hamiltonian flow (projectivised Schrödinger)	3.6
μ_L	Liouville measure on Σ	2.4
μ_{FS}	Fubini–Study measure on $\mathbb{CP}^{N-1}$	3.3
$\omega_\Sigma, \omega_{FS}$	Symplectic forms (always subscripted)	2.3
$\hat{H}, h, H$	Operator, epistemic Hamiltonian, ontic Hamiltonian	3.7, 3.6, 2.5
$\rho_{ep}, \hat{\rho}$	Epistemic density, density operator	3.4, 3.8
$\Omega_i^\Sigma, \Omega_i^{ep}$	Outcome regions (ontic and epistemic)	2.7, 3.5
w_i	Outcome weight (probability) for outcome i in a fixed measurement context M : $w_i(M; \nu_{ep}) := \nu_{ep}(\Omega_i^{ep}(M)) = \int_{\Omega_i^{ep}(M)} \rho_{ep} d\mu_{FS}$ (equivalently $w_i = \text{Tr}(\hat{\rho} \Pi_i)$ for projective M).	§ 3.9